

Table A.1. BSRN station-to-archive file format update LR4000 – pyrgeometer temperatures

Logical record	Line no.	Description of field / format of line	Range of values	Missing code	Format of v./l.
4000 pyrgeo. temp	1	date [day]	1-31		I2
	1	time [minute]	0-1439		I4
	1	dome temperature 1 downward long-wave instrument [°C]		-99.99	F6.2
	1	dome temperature 2 downward long-wave instrument [°C]		-99.99	F6.2
	1	dome temperature 3 downward long-wave instrument [°C]		-99.99	F6.2
	1	body temperature downward long-wave instrument [°C]		-99.99	F6.2
	1	thermopile output downward long-wave instrument [W/m²]		-999.9	F6.1
	1	dome temperature 1 upward long-wave instrument [°C]		-99.99	F6.2
	1	dome temperature 2 upward long-wave instrument [°C]		-99.99	F6.2
	1	dome temperature 3 upward long-wave instrument [°C]		-99.99	F6.2
	1	body temperature upward long-wave instrument [°C]		-99.99	F6.2
	1	thermopile output upward long-wave instrument [W/m²]		-999.9	F6.1
		(X,I2,X,I4,X,4(F6.2,X),F6.1,2X,4(F6.2,X),F6.1)			
4nnn pyrgeo. temp. at nnn meter		Pyrgeometer temperatures from instruments mounted on towers at a height of nnn meters are coded according to the definitions for pyrgeometers at a standard height (~2 m), see LR4000			

Note: the thermopile outputs are defined as U(output voltage of the thermopile)/C (sensitivity of the pyrgeometer)

Any station-to-archive files containing a LR4000 (or LR4nnn) should contain also a metadata line, for each pyrgeometer, incorporated in LR0003 (commentary section). Each line, starting with the meta-code @LR4000CONST (or @LR4nnnCONST), should contain the information about the pyrgeometer IDs (manufacturer and WMO), calibration certificate id, and correction parameters as defined by the following template and general equation of the pyranometer:

@LR4000CONST, s/n (Manufacturer), s/n (WMO), CertificateCodeID¹, C, k0, k1, k2, k3, f

$$L = k_0 + \frac{U_{emf}}{C} (1 + k_1 \sigma T_B^3) + k_2 \sigma T_B^4 - k_3 \sigma (T_D^4 - T_B^4) - f \Delta T_{N-S}$$

By setting any missing information with the “ND” value.

Example:

@LR4000CONST, 040736, 53005, 2015-1064-02, 12.25, ND, 0.01, 0.9994, ND, ND

@LR4000CONST, 170259, 53019, 2019-1769-01, 9.97, ND, 0.03, 0.9989, ND, ND

Update date: 2022-08-17

¹ Please send a copy of your calibration certificates to BSRN PM or WRMC director or upload them in your BSRN calibration folder.